



## BACHELOR OF COMPUTER SCIENCE (HONS)(SOFTWARE ENGINEERING)

### TAKE HOME FINAL EXAMINATION JANUARY 2020

Course : EC3307 (Object and Component Technology) Duration to complete: 72 Hours  
Start Time: 1.30 pm, 11 May 2020  
Lecturer : Ms.Rajesvary Rajoo End Time : 1.30 pm, 14 May 2020

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#### **Instructions:**

Answer **ALL** questions.

This examination paper is confidential. The questions must be answered individually. Students are NOT PERMITTED to discuss or consult with other students or individuals.

**Permitted reference materials:** Lecture notes, text book

Reminder: references must be cited accordingly.

#### **Answer Format:**

- i. Do not put your name on any materials related to the exam. Use only your Student ID Number for identification.
- ii. Your answer can be submitted via a word document.
- iii. Diagrams can be submitted via picture.
- iv. Please save your answers in the following format:  
**STUDENT ID\_COURSE CODE\_COURSE TITLE\_LLECTURER NAME**
- v. Your answer file must reach my email ([rajes\\_e@nilai.edu.my](mailto:rajes_e@nilai.edu.my)) by 1.30 pm, 14<sup>th</sup> May 2020

#### **Honor Pledge for Exams**

"I affirm that I have not given or received any unauthorised help on this exam, and that all work is my own."

Name and Signature: \_\_\_\_\_

This question paper consists of 4 pages.

1. Consider the below specification of an abstract data type (ADT) Dictionary, that relates integer keys to string values. It is defined as:

```
public interface Dictionary
{
    public int capacity(Dictionary d);
    /*      pre: true
       post: capacity(d) = 'the maximum size of the Dictionary d' */

    public int size(Dictionary d);
    /*      pre: true
       post: size(d) = 'the (current) size of the Dictionary d' */

    boolean isln(Dictionary d; int x);
    /*      pre: true
       post: isln(d, x) = 'an element with key x is in the Dictionary d' */

    public void put( Dictionary d; int x; String v);
    /*      pre: size(d) <> capacity(d) and not isln(d, x)
       post: isln(d, x) and (size(d) = size(old d) + 1)
       note: old d denotes the value of d before the procedure */

    public void get(Dictionary d; int x; String v);
    /*      pre: isln(d, x)
       post: v is value at key x in Dictionary d */

    public void remove(Dictionary d; int x);
    /*      pre: isln(d, x)
       post: not isln(d, x) and (size(d) = size(old d) - 1) */
}
```

- a. How can you relate the precondition and post-condition of Hoare logic and the contract of Meyer's 'design by contract' to method put () for the ADT Dictionary. **(6 MARKS)**
- b. How do you classify the methods of Dictionary into basic queries, derived queries and operations? Explain your choices. **(6 MARKS)**
- c. Explain how a client (user) of the abstract data type Dictionary could ensure that a call of method put conforms to the 'contract' implied by the specification of Dictionary. What assumption you can make? **(5 MARKS)**
- d. Assuming that a supplier implements the method get () as follows: **(5 MARKS)**

```

public void get ( Dictionary d; int x; String v); {
    /*      pre: isln(d, x)
    post: v is value at key x in Dictionary d */

    if isln(d, x) then
        /* statements to get element at x from Dictionary d into v
    */

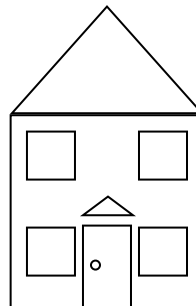
    else v = " " /* empty string */
}

```

Bearing in mind the behaviour of method get () when the pre-condition is not satisfied, does this implementation conform to its specification? Explain your answer.

- e. Explain the consequences of the assumption that a pre-condition of method get() is false. (3 MARKS)

2. A composite design pattern can be used to design a system that can represent and render (draw on the screen) graphical objects composed of multiple points, lines, ellipses, rectangles and triangles. An example of the kind of graphical object to represent and render is given below:



The following are some examples of the methods the system can provide:

```

public void Draw(Window w)
Renders the object in window w.

```

```

public void Translate(int x_trans, int y_trans )
The value of x_trans is added to the x coordintate(s) of the object. The value of
y_trans is added to the y coordinate(s) of the object.

```

```

public void Scale(int s)
The x and y coordinate(s) of the object are multiplied by s.

```

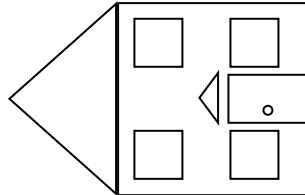
```

public void Rotate(Point p, double thisAngle)

```

The object is rotated about point p by thisAngle.

For example, if the client had created a graphic corresponding to the house diagram above, then applying a rotation of 90 degrees about centre of the house should end up with the graphic shown below (note that the client calls the rotate command on the entire graphic, your system will take care of how to rotate each of the components of the graphic):



- a. Give an example application that suit for Composite design pattern as described above. Explain your choice. **(5 MARKS)**
- b. Draw the class diagram to represent the example in (a). Identify the attributes and methods to be defined in each class. Explain briefly the attributes and methods that are suggested by the design pattern. **(9 MARKS)**
- c. An additional feature is required of the graphics system that will allow the client to display some new features around an existing individual or composite graphic object.
  - i. Choose a design pattern with which to implement the decorated feature of the graphic system. Explain your choice **(3 MARKS)**
  - ii. Redraw your class hierarchy, including any new classes for the additional features which are required by the design pattern. Identify the attributes and methods to be defined in each class and explain the attributes and methods that are suggested by the design pattern you have used. **(8 MARKS)**

3. a. Implement the `Addable<E>` and `Stats<E extends Number>` (10 MARKS)  
interfaces, and the `StatsColl<E extends Number>` class, which  
were introduced in the lecture.
- b. Use the `Scanner` class to implement a code which meets the (10 MARKS)  
following requirement:  
It maintains two collections, one of type  
`StatsColl<Integer>` and the other of type  
`StatsColl<Double>`.
- c. Give an example of how a `ClassCastException` may be thrown in (5 MARKS)  
Java when retrieving an item from an `ArrayList`.

4. As the time is changing, technologies are expanding and with that, the priority and the importance of the technologies for application development. It becomes difficult to pick when we have to choose among major different technologies. Java and Microsoft .NET are two leading technologies intended for development of desktop and server-side applications. Both platforms enable the use of high-level programming languages to build large-scale applications. You can build excellent software on both .NET and Java.

Java was introduced by Sun, and of course it wanted to control the evolution. However, Microsoft also wanted to change Java in certain ways to suit its requirements. Considering the legal issues, Microsoft decided to design its own language, which is C#. C# is Microsoft's preferred language that best supports .NET and is best supported by .NET. Therefore, there are some similarities and differences between Java and C#. One of it is goto statement. The much maligned goto statement is re-introduced in C#, admittedly with restrictions so that it is scarcely possible to misuse it.

- a. Give an example application and state whether you would choose (3 MARKS)  
Java technology or .Net technology as a platform for development  
of the application.
- b. By using the example application in (a), explain in detail **any FOUR** (12 MARKS)  
(4) deciding factors for choosing the particular technology as  
compared to the other. (Note: your explanation should be precise  
and detail)
- c. Do you agree that the limited reintroduction of the goto is not (10 MARKS)  
considered harmful if used wisely? Give a simple code example to  
illustrate your answer. Otherwise, give a counter argument with an  
example code.

-END OF QUESTION PAPER-